

A Social Media Database System

Minor Project Proposal

CS 257: Database and Information Systems Lab

Team Members

Arnav Kumar	240001013
Aryaman Tiwari	240001014
Ayush Singh Rana	240001015
Hrishabh Mittal	240001035

Contents

1	Problem Implementation	1
2	Technology Stack	2
2.1	Frontend	2
2.2	Backend	2
2.3	Database	2
2.4	Real-time & Integration	2

1 Problem Implementation

The proposed project implements a **social media database system** designed to manage complex social interactions, multimedia content, and user engagement while ensuring scalability, transparency, and efficient data handling.

At the foundation of the system are three **strong entities**:

- **User**: Represents the core identity of each participant, storing login credentials, personal details, profiles, and customizable privacy settings.
- **Post**: Encapsulates user-generated content, including text, media, timestamps, and visibility settings.
- **Comment**: Enables threaded discussions by linking user input to posts and supporting hierarchical replies via parent-child comment structures.

In addition, two **weak entities** capture dependent relationships:

- **Reaction**: Represents user engagement with posts. Each reaction is uniquely identified by the combination of a user and a post, along with the reaction type (e.g., like, dislike, emoji).
- **Message**: Models direct communication between two users. Since its identity depends on the interacting pair and the timestamp, it is treated as a weak entity.

To capture the complexity of social interactions, two **associative entities** are introduced:

- **Friendship**: Represents a mutual relationship between two users. It stores metrics such as number of exchanged messages, likes, and comments, which are combined into a computed *Friendship Score* for ranking and recommendation purposes.
- **Following**: Captures one-directional connections, storing status and timestamp to differentiate between active and pending follow requests.

The system's relationships reflect real-world behaviors on social networks. Users can *create* posts and *make* comments, while posts themselves *have* comments to enable rich interaction threads. Users can *give* reactions to posts, linking engagement directly to content, and they can *send* messages to each other, forming conversations. Friendships and followings together model both mutual and one-way social bonds.

In summary, these entities and relationships form a tightly interconnected structure that mirrors real-world online communities, ensuring support for user identity, social connectivity, and interactive engagement.

2 Technology Stack

2.1 Frontend

- **Vite**: Fast build tool and development server
- **HTML/CSS**: Semantic markup and styling
- **Tailwind CSS**: Utility-first CSS framework for responsive design
- **React**: Component-based UI library for interactive interfaces

2.2 Backend

- **Node.js**: JavaScript runtime for server-side development
- **Express.js**: Web application framework for API development

2.3 Database

- **PostgreSQL**: Primary relational database for structured data
- **Redis**: In-memory cache for session management and real-time features
- **Prisma**: Database ORM for type-safe database access

2.4 Real-time & Integration

- **Socket.io**: WebSocket implementation for real-time messaging
- **OAuth**: Third-party authentication integration

This project shows how a social media platform can be built around clear and reliable data management. By focusing on users, their connections, and the content they share, the system supports smooth interactions, quick access to information, and meaningful recommendations. The design ensures that conversations, posts, and friendships all work together in a way that feels natural and reflects the complexity of real online communities.